

Preamp Overload



Duncanamps

Site: **Duncanamps**

Saved Today at 21:40

...

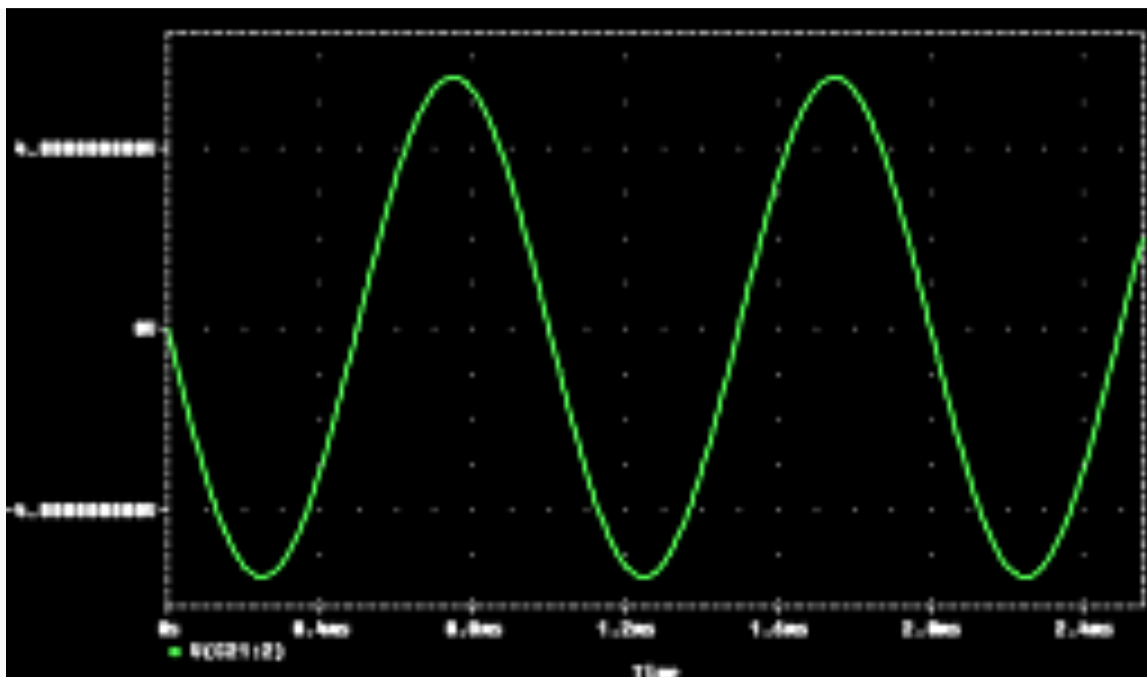
Ever hear notes disappear on a valve amp? You pick the note - then some silence, then the note fades in. A bit like violin techniques, but you get to hear the initial pick of the note.

This is normally caused by overloading one or more of the preamp stages.

Shown to the right is a preamp stage. OK, they may vary a bit, but this is a normal bargain basement tube preamp stage as used in a guitar amp.

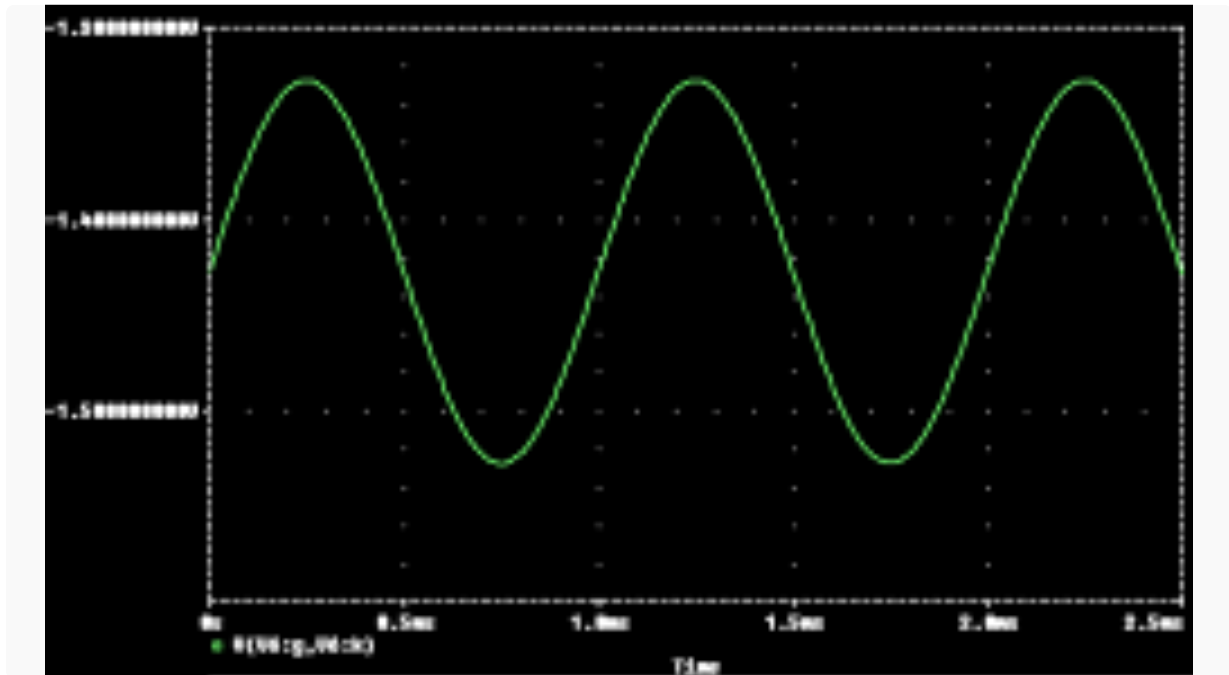
The input is coupled to the grid of the valve using a capacitor. It may have come from the anode of a previous preamp valve. A 1 meg resistor connects the grid to ground, so the most of the time, the grid will be at zero volts.

The 1.8k resistor on the cathode allows the cathode to float up to, say, a couple of volts. This is chosen to give the right amount of bias for the valve, so that the anode floats at approximately half of the B+ voltage.



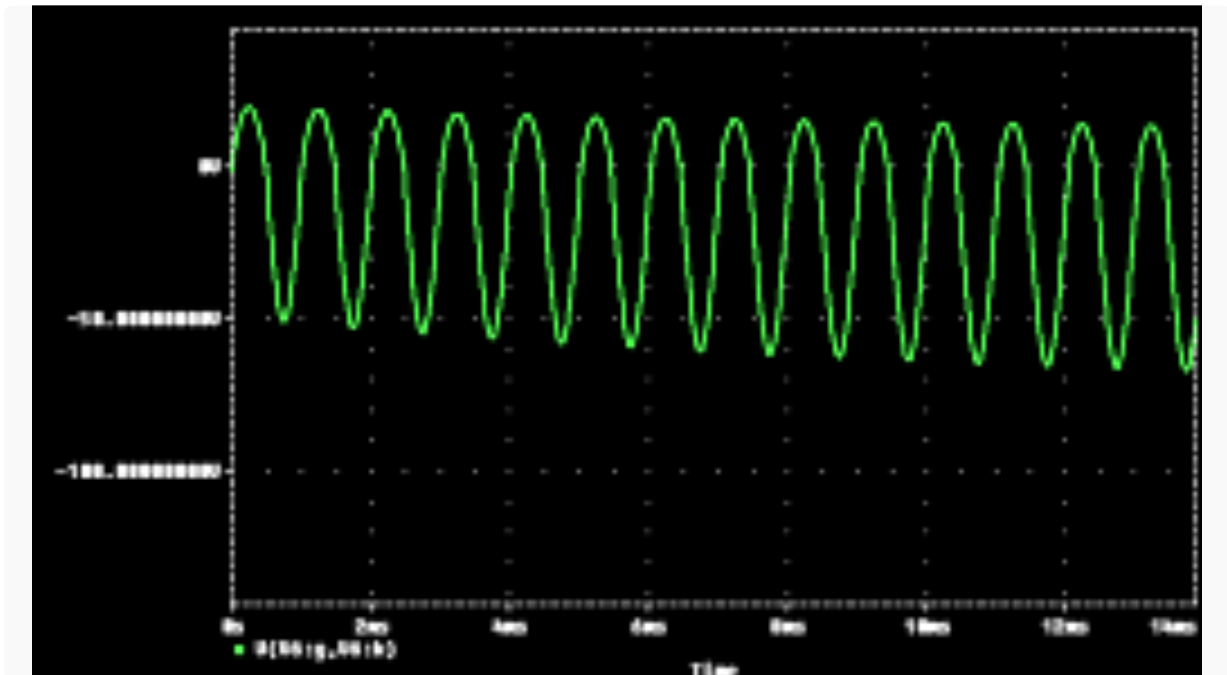
OK, so we feed in 200mV peak to peak sine wave at the input terminal. Check the trace on the right, we get 11V peak to peak output, a gain of just over 50.

ps: you can **click on any of the traces** to see them full size. The output looks nice and clean, because the preamp is being operated well within its signal handling limits.



So what happens if we try and overload the preamp, and what is the effect of overload. Check the grid to cathode voltage on the next trace. This is the voltage on the grid relative to the cathode, and hovers around -1.4 volts.

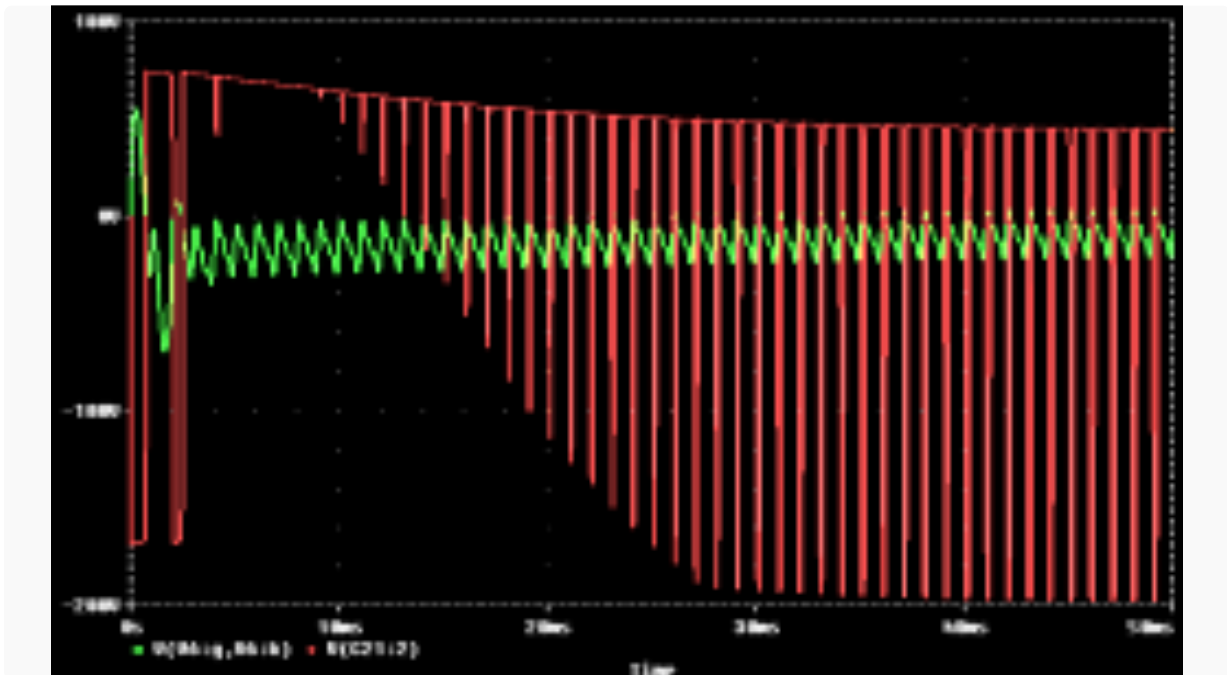
If we push the input signal up, at a certain point the valve will start to draw grid current. This will happen when the grid to cathode voltage is around 0 volts, but will vary slightly from valve to valve.



If we feed a signal of 100 volts peak to peak into the preamp, the grid to cathode voltage should vary between -51.4 volts and +48.6V (that's the theory, anyway).

Watch what happens to the grid to cathode voltage... The tops are effectively chopped off as the valve goes into grid current. Notice also how the lower end of the waveform is dropping away, as that coupling capacitor starts to charge up.

This is all somewhat abstract, as guitarists don't feed clean sine waves into their amplifiers! A guitar signal consists of an initial pick, followed by a decaying waveform with some harmonics.



Check what happens in the trace on the right when we feed a guitar signal in with a total amplitude of 100 volts peak to peak.

Superimposed on the green trace (the guitar signal), is a red trace which shows the OUTPUT from this preamp stage. Note the initial pick produces a nasty distorted blip, followed by silence until the input capacitor discharges back to a state which does not cause the valve to get cut off.

How do we fix this ?

The culprit in all of this is the input capacitor. By using the initial pick and the grid current of the valve to charge it up, we end up with a situation where the valve is cut off for a short period of time.

There are a number of approaches which can be taken:

1. Leave it alone - some people like this effect!
2. Reduce the B+ supply to the previous stage - the lower the amount of drive, the less chance of cutting off the valve.
3. Reduce the drive from the previous stage by means of cascaded resistors (see pic below).
4. Symetrically clip the input to the grid - use a bridge rectifier "Marshall style".

Caveats...

There are some caveats with all of this.... Reducing the B+ will reduce gain slightly, but will reduce headroom by a large amount too. If clean country picking is your chosen style, then taking the B+ down will give you some nice distorted country picking....

Cascaded resistors will reduce gain. In the example below, the gain is reduced by a factor of six. This will make the overall stage gain around 10 instead of 50, so you will need more valves to accomplish the same amount of gain.

Using a bridge rectifier to clip the input will lead to a very, symmetric, erm, Marshall sound. This may not be what you want. For my own purposes, I tend to use cascading resistors to reduce gain by a factor of 3, and run from a B+ of 56 volts. These will allow you (with sufficient valves) to go from a nice clean sound, to metal madness....

Ah yes, that picture:

Note how the anode load has been split - this cuts down the amount of drive to the next stage, and helps prevent cutoff.

The traces shown above are computer simulations rather than real information taken from an oscilloscope. The modelling was carried out using PSpice from Orcad.